

Priyanshu Pratik

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EDUCATION

Gati Shakti Vishwavidyalaya

Bachelor of Technology in Artificial Intelligence and Data Science

Vadodara, India

Exp. 2028

Specialisation: Transportation and Logistics

- Current Status: 2nd Year (4th Semester) | CGPA: 8.5+

- Relevant Coursework: Machine Learning, Pattern Recognition, Design and Analysis of Algorithms, Signals and Systems, Computer Organisation and Architecture.

Delhi Public School

Class XII (Higher Secondary)

Vadodara, India

Year 2023

- Percentage: 87.0% (434/500)

- Joint Entrance Examination (JEE) Main: 96.6 Percentile (AIR: ~50,000)

Soar Valley College

Class X (Secondary / GCSE)

Leicester, United Kingdom

Year 2020

TECHNICAL SKILLS

Languages: Python, Kotlin, Java, Javascript, C, MIPS Assembly, SQL

AI & Data Science: Machine Learning, Computer Vision, NLP, Pandas, PyTorch, TensorFlow Lite

Tools & Frameworks: Jetpack Compose, Streamlit, PyQt6, Git/GitHub, Docker

Core Concepts: Data Structures & Algorithms, DSP (FFT), MVVM Architecture, Object-Oriented Programming

PROJECTS

PulmoSense (Audio Breathing Analysis) | Kotlin, PyTorch, TFLite, Audio Processing

Apr 2026

- Architected an AI-driven Android application utilizing a custom TensorFlow Lite model to analyze respiratory audio patterns and deliver real-time health insights.
- Deployed a custom processing engine (PulmoEngine) to compute 128-band Mel spectrograms from raw recordings, optimizing for real-time edge inference.

Aura (Real-Time Audio Identifier) | Kotlin, Jetpack Compose, DSP

Mar 2026 – Present

- Engineered a fault-tolerant Android audio identifier capable of real-time acoustic fingerprinting and "Query by Humming" melody matching using Fast Fourier Transform (FFT) algorithms.
- Designed a reactive Jetpack Compose UI following MVVM architecture, integrating intelligent metadata parsing and platform-specific deep-linking for seamless song accessibility.

Waveglider (3D Flight Simulation) | Web Technologies, Javascript, 3D Rendering

Feb 2026 – Present

- Designed an interactive, web-based 3D flight simulation featuring a user-controlled entity navigating an infinite, procedurally generated ocean environment.
- Optimized real-time rendering and dynamic third-person camera tracking to deliver smooth, low-latency keyboard control mechanics.

LocalPDF Pro (Desktop & Mobile Utility) | Python, Kotlin, PyQt6

Dec 2025 – Present

- Deployed an open-source, 100% offline desktop application and corresponding Kotlin Android utility, providing secure, native PDF processing without cloud dependency.
- Integrated an interactive analytics dashboard to track processed data volumes, prioritizing user privacy and standalone execution.

Wayfinder (Pathfinding Visualiser) | Javascript, Algorithmic Routing

Dec 2025 – Present

- Designed an interactive visualization platform for complex routing algorithms (A*, Dijkstra's), specifically optimized to model spatial traversal problems.
- Demonstrated efficient map-based routing solutions directly applicable to large-scale transportation and logistics mapping systems.

Beyond the Apex (F1 Telemetry Engine) | Python, Javascript, FastF1 API, Data Visualisation

Nov 2025 – Present

- A monetized telemetry analysis platform utilizing the FastF1 API, enabling users to generate and share 2D comparative visualizations for multi-driver session analysis under a tiered subscription model.

PROBLEM SOLVING & ACADEMIC ENDEAVORS

- **Competitive Programming & Cryptic Hunts:** Solved 160+ algorithmic problems on LeetCode and actively participate in complex puzzle-solving events (e.g., Cipher Saga) requiring rigorous analytical thinking.
- **GATE 2027 Aspirant:** Actively preparing for the GATE Data Science and Artificial Intelligence (DA) paper, demonstrating a commitment to rigorous academic standards in core AI domains.

INTERESTS

Space Exploration, Aeronautics, Astrophysics, Telemetry Analysis, Game Development.